PEMRT525 — IoT & Applications

Viva Question Bank with Answers

S5 Mechatronics (MR 2K24) | JECC | Faculty: Mr. Jinesh K J

4 Modules · 10 Questions Each · 40 Questions Total

MODULE 1 — Introduction to IoT

Q1. What is IoT? Give its full form.

IoT stands for **Internet of Things**. It is a network of physical objects embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet.

Q2. List the five key characteristics of IoT (DSIUI).

- Dynamic & self-adapting
- Self-configuring
- Interoperable
- Unique identity (IP address / URI)
- Integrated into the information network

Q3. What are the two components of IoT Physical Design?

- **Things** — physical devices with sensing/actuation capability
- **Protocol Stack** — communication layers (application, transport, network, link, physical)

Q4. Name the functional blocks in IoT Logical Design.

- Device
- Communication
- Services
- Management
- Security
- Application

Q5. Name the four IoT communication models.

- Request-Response (HTTP, CoAP)
- Publish-Subscribe (MQTT)
- Push-Pull (queue-based)
- Exclusive Pair (WebSocket)

Q6. Explain the Publish-Subscribe communication model.

A **broker** sits between publishers and subscribers. Publishers send messages to a topic; subscribers receive all messages on that topic. Neither knows the other directly. Used in **MQTT**.

Q7. What is the Exclusive Pair communication model?

A **persistent, full-duplex** connection between client and server. Both can send data at any time without a new request. **WebSocket** uses this model.

Q8. What are IoT enabling technologies? Name any four.

- Wireless Sensor Networks (WSN)
- Cloud Computing
- Embedded Systems
- RFID
- Big Data Analytics
- Communication Protocols (MQTT, CoAP)

Q9. What is IoT Level 1? Give its characteristics.

The simplest IoT configuration — a **single node** that senses, processes data locally, and controls an actuator. No cloud or internet required. Example: a thermostat that reads temperature and turns on a fan.

Q10. Difference between IoT Level 1 and Level 6?

Level 1 — single node, local analysis, no cloud.

Level 6 — multiple nodes, cloud storage, big data analytics, remote monitoring, full internet integration.

MODULE 2 — M2M, SDN/NFV, Smart Objects & Sensor Networks

Q1. What is M2M communication?

Machine-to-Machine (M2M) is direct communication between devices for remote monitoring and control without human intervention. Example: a vending machine automatically reporting low stock to a server.

Q2. Key differences between M2M and IoT.

Aspect	M2M	IoT
Protocols	Proprietary / non-IP	Open IP-based (HTTP, MQTT)
Connectivity	Closed, point-to-point	Open internet
Data storage	On-premises (siloed)	Cloud
Scalability	Limited	Highly scalable

Q3. What is SDN? What problem does it solve?

Software Defined Networking (SDN) separates the **control plane** (routing decisions) from the **data plane** (packet forwarding) and centralises control in a software controller. Solves the complexity of managing large networks.

Q4. Name the three layers of SDN.

- **Infrastructure Layer (Data Plane)** — switches/routers that forward packets
- **Control Layer (Control Plane)** — centralised SDN controller (the "brain")
- **Application Layer** — SDN apps via northbound APIs

Q5. What is OpenFlow in SDN?

OpenFlow is the standard southbound protocol between the SDN controller and network switches. The controller programs flow tables in switches to control packet forwarding.

Q6. What is NFV? How does it differ from SDN?

NFV runs network functions (firewalls, NAT, load balancers) as software on standard servers.

SDN = centralises control of traffic. **NFV** = virtualises the network functions themselves. They complement each other.

Q7. What are the three components of NFV architecture?

- **VNF** — Virtualized Network Function
- **NFVI** — NFV Infrastructure (servers, storage, network)
- **NFV MANO** — Management & Orchestration

Q8. What is a sensor? What is an actuator? Give one example each.

Sensor — measures a physical quantity and converts to electrical signal. Example: DHT22.

Actuator — converts electrical signal into physical action. Example: Solenoid valve, relay.

Q9. What is MEMS? Where is it used in IoT?

MEMS (Micro-Electro-Mechanical Systems) are miniaturised devices integrating mechanical and electrical components on a chip. Used as accelerometers, gyroscopes, pressure sensors in IoT devices.

Q10. What is a WSN? What are the communication criteria for connecting smart objects?

A **WSN** is a network of spatially distributed sensor nodes that wirelessly communicate with a gateway.

Communication criteria: **Range**, **Power consumption**, **Bandwidth/data rate**, **Reliability/QoS**, **Cost**.

MODULE 3 — IoT Communication Protocols

Q1. What is IEEE 802.15.4? What does it define?

IEEE 802.15.4 defines the **Physical (PHY) layer** and **MAC layer** for low-rate Wireless Personal Area Networks (LR-WPANs). It is the foundation used by ZigBee, 6LoWPAN, and other IoT standards.

Q2. What does the IEEE 802.15.4 Physical Layer do?

- Radio frequency operation (868 MHz, 915 MHz, 2.4 GHz bands)
- Modulation using DSSS (Direct Sequence Spread Spectrum)
- Clear Channel Assessment (CCA) — checks if channel is free before transmitting

Q3. What does the IEEE 802.15.4 MAC Layer do?

- **CSMA-CA** — manages channel access (avoids collisions)
- Beacon generation and synchronisation
- Association and disassociation of devices
- Reliable link using acknowledgements (ACK)

Q4. What are 802.15.4e and 802.15.4g?

802.15.4e — enhanced MAC amendment for industrial IoT; adds TSCH (Time Slotted Channel Hopping) for reliability in factory automation.

802.15.4g — PHY amendment for Smart Utility Networks; adds sub-GHz bands and higher data rates for smart grid / metering.

Q5. What is Modbus? Name its three variants.

Modbus is a serial communication protocol for industrial devices (developed 1979).

- **Modbus RTU** — binary encoding, compact, RS-485
- **Modbus ASCII** — ASCII encoding, human-readable
- **Modbus TCP** — Modbus over Ethernet/TCP-IP

Q6. Name the four register types in Modbus.

- **Coils** — read/write, 1-bit (digital output)
- **Discrete Inputs** — read-only, 1-bit (digital input)
- **Holding Registers** — read/write, 16-bit (analog output / config)
- **Input Registers** — read-only, 16-bit (sensor data)

Q7. What are the three ZigBee device types and topologies?

Device types: **Coordinator** (manages network), **Router** (extends range), **End Device** (sleeps to save power).

Topologies: **Star** (all to coordinator), **Tree** (hierarchical), **Mesh** (any router to any router — most robust, uses AODV routing).

Q8. What modulation does LoRaWAN use? Explain Spreading Factors.

CSS (Chirp Spread Spectrum) — frequency sweeps over time. Gives long range + interference resistance.

SF7–SF12: **SF7** = highest data rate, shortest range. **SF12** = lowest data rate, longest range. Higher SF = longer range but slower.

Q9. What is LTE-M? How does it differ from NB-IoT?

Feature	LTE-M	NB-IoT
Bandwidth	1.4 MHz	180 kHz
Data rate	~1 Mbps	~250 kbps
Mobility	Supports handover	Limited
Voice	Supports VoLTE	No
Use case	Asset tracking, wearables	Smart meters, sensors

Q10. What is NB-IoT? Give its key specifications.

NB-IoT (Narrowband IoT) — 3GPP Release 13 LPWA standard:

- Bandwidth: **180 kHz** (one LTE resource block)
- Operates in **licensed cellular bands**
- Range: up to 10 km; excellent deep indoor penetration
- Ultra-low power — battery can last 10+ years
- Uses existing cellular towers (no new infrastructure)

MODULE 4 — RPi Programming, IoT Security & Smart City

Q1. What is the Raspberry Pi GPIO header? How many pins?

GPIO (General Purpose Input/Output) is the **40-pin header** on the Raspberry Pi providing power pins (3.3V, 5V), ground pins, and programmable I/O pins for sensors, actuators, and displays.

Q2. Difference between BCM and BOARD pin numbering in RPi.GPIO?

BCM — Broadcom chip GPIO numbering (e.g., GPIO17). `GPIO.setmode(GPIO.BCM)`

BOARD — physical pin position on 40-pin header (e.g., pin 11). `GPIO.setmode(GPIO.BOARD)`

Q3. Communication interfaces on RPi and their BCM pin numbers.

Interface	Pins (BCM)	Python Library
UART	GPIO 14 (TX), GPIO 15 (RX)	pyserial
I2C	GPIO 2 (SDA), GPIO 3 (SCL)	smbus2
SPI	GPIO10(MOSI) GPIO9(MISO) GPIO11(SCLK) GPIO8(CE)	spidev

Q4. Write Python code to blink an LED on GPIO 17 (BCM).

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(17, GPIO.OUT)
for _ in range(5):
    GPIO.output(17, GPIO.HIGH)
    time.sleep(1)
    GPIO.output(17, GPIO.LOW)
    time.sleep(1)
GPIO.cleanup()
```

Q5. What is Wokwi? How is it useful?

Wokwi (wokwi.com) is a free browser-based IoT simulator for Arduino, ESP32, and Raspberry Pi. Build circuits and run Python/C++ code without physical hardware — useful for learning and testing.

Q6. What is OWASP IoT Top 10? Name the first three vulnerabilities.

OWASP IoT Top 10 lists the most critical IoT security risks:

- Weak, guessable, or hardcoded passwords
- Insecure network services
- Insecure ecosystem interfaces (APIs, cloud, mobile)

Q7. What is the STRIDE threat model?

- **S** — Spoofing (pretending to be someone else)
- **T** — Tampering (modifying data)
- **R** — Repudiation (denying an action)
- **I** — Information Disclosure (data leak)
- **D** — Denial of Service
- **E** — Elevation of Privilege

Q8. What is the Layered Attacker Model in IoT security?

Classifies attackers by the OSI layer they target:

- **Physical** — hardware tampering, device cloning
- **Data Link** — jamming, eavesdropping
- **Network** — IP spoofing, routing attacks
- **Transport** — TCP/UDP flooding
- **Application** — malware injection, software exploits
- **Cross-layer** — combined attacks across multiple layers

Q9. What is Secure Message Communication in IoT? Key properties?

- **Confidentiality** — encryption prevents unauthorised reading (AES, TLS)
- **Integrity** — HMAC ensures data is not altered in transit
- **Non-repudiation** — digital signatures prove sender identity
- **Availability** — protection against DoS attacks

Protocols: **TLS/DTLS**, **MQTT with TLS**, **CoAP with DTLS**.

Q10. Describe the 4-Layer Smart City Architecture. Fog vs Cloud?

- **Street Layer** — sensors, cameras, meters collect raw data
- **City Layer** — Wi-Fi, 4G/5G, fibre transmits data
- **Data Center Layer** — cloud/fog servers store and process data
- **Services Layer** — citizen apps: traffic, parking, alerts

Fog Computing — processes data near the device (edge), low latency, real-time.

Cloud Computing — processes in remote data centres, high capacity, big data analytics.